Problem Set 5A

Vector equations of lines

Do these after the slides for Section 5A. Textbook Exercise 5A. Answers are on the last page. Show full working in your exercise book.

Skills

1 Write a vector equation of the line through $(3, -1, 2)$ parallel to $\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$.

2 For the line through $A(1, 0, -2)$ and $B(4, 2, 3)$, write:

(a) a vector equation

(b) parametric equations

(c) a Cartesian equation

3 For the line in Question 2, decide whether each point lies on the line.

(a) $P(7, 4, 8)$

(b) $Q(-2, -2, -6)$

4 A line has Cartesian equation $\dfrac{x-2}{4} = \dfrac{y+1}{3} = \dfrac{z}{-2}$.

(a) Write the line in vector form.

(b) Find where the line meets the yz-plane.

5 Find the distance from $P(5, 0, 4)$ to the line $\mathbf{r} = 2\mathbf{i} + \mathbf{k} + t(\mathbf{i} + 2\mathbf{j} + 2\mathbf{k})$.

Exam style

6 Find the coordinates of the points on the line $\mathbf{r} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k} + t(2\mathbf{i} - 2\mathbf{j} + \mathbf{k})$ that are 12 units from $(2, 1, -3)$.

7 Find the point on the line $\mathbf{r} = 5\mathbf{i} + \mathbf{j} + \mathbf{k} + t(\mathbf{i} + 2\mathbf{j} + 2\mathbf{k})$ that is closest to the origin, and the distance from the origin to the line.

8 The line ℓ passes through $(1, 3, -2)$ and is parallel to the line $\dfrac{x-1}{2} = \dfrac{y+4}{-1} = \dfrac{z}{3}$. Find where ℓ crosses the plane $z = 4$.

Problem Set 5A: answers

1 $\mathbf{r} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k} + t(\mathbf{i} + 4\mathbf{j} - 2\mathbf{k})$

2 (a) $\mathbf{r} = \mathbf{i} - 2\mathbf{k} + t(3\mathbf{i} + 2\mathbf{j} + 5\mathbf{k})$ (b) $x = 1 + 3t,\ y = 2t,\ z = -2 + 5t$ (c) $\dfrac{x-1}{3} = \dfrac{y}{2} = \dfrac{z+2}{5}$

3 (a) Yes ($t = 2$) (b) No ($t = -1$ gives $z = -7$)

4 (a) $\mathbf{r} = 2\mathbf{i} - \mathbf{j} + t(4\mathbf{i} + 3\mathbf{j} - 2\mathbf{k})$ (b) $\left(0, -\frac{5}{2}, 1\right)$

5 Closest point $(3, 2, 3)$ at $t = 1$; distance 3

6 $t = \pm 4$: $(10, -7, 1)$ and $(-6, 9, -7)$

7 $\mathbf{r} \cdot \mathbf{d} = 9 + 9t = 0$, $t = -1$: point $(4, -1, -1)$, distance $3\sqrt{2}$

8 $t = 2$: $(5, 1, 4)$